

2019 Ph H1 Q14

Section: Particles and Waves

Topic: Nuclear Reactions

Question Summary

Francium-223 undergoes β^- decay: ${}^{223}_{87}\text{Fr} \rightarrow Y + \beta^-$. Identify Y.

Worked Solution

In β^- decay, A stays constant; Z increases by 1.

223 remains 223; $87 \rightarrow 88$ (radium).

Final Answer

C — ${}^{223}_{88}\text{Ra}$

Revision Tips

- In β^- decay: A constant; Z increases by 1.
- Neutron $\rightarrow$ proton + electron + antineutrino.
- Use periodic table to match Z.